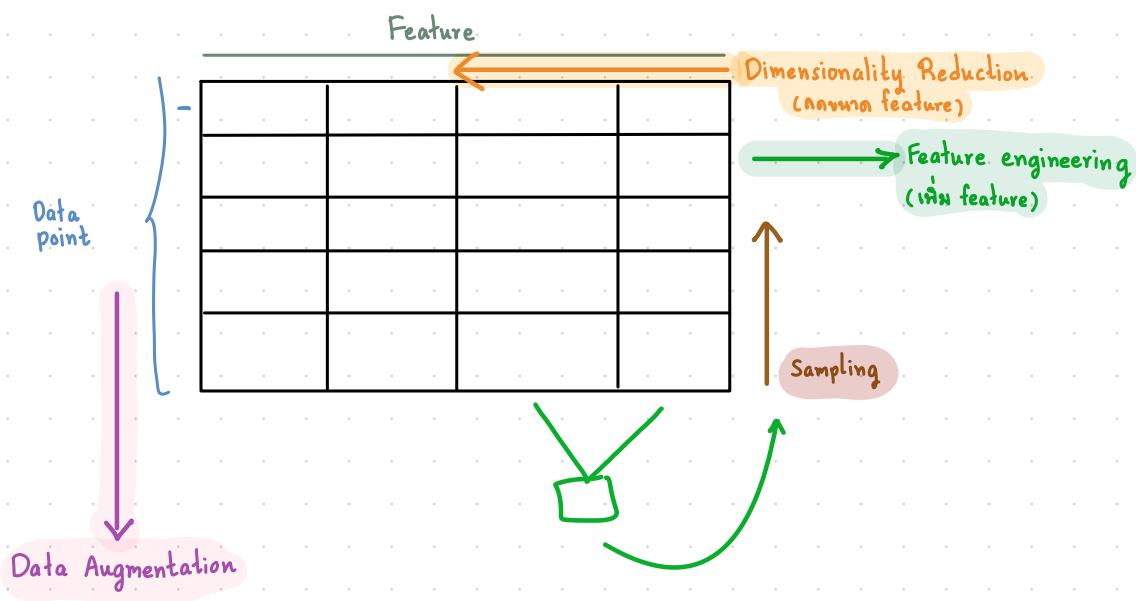


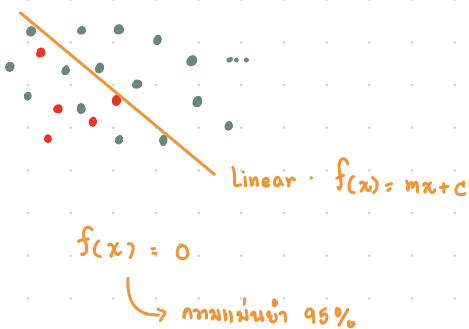
Dimensionality Reduction Modification



denote as

- sick (5) = 1
- normal (95) = 0

Occam's razor
เลือกวิธีที่ง่ายที่สุด
ก่อนพิจารณาวิธีที่ซับซ้อน



denote as

- sick (5) = 1
- normal (95) = 0

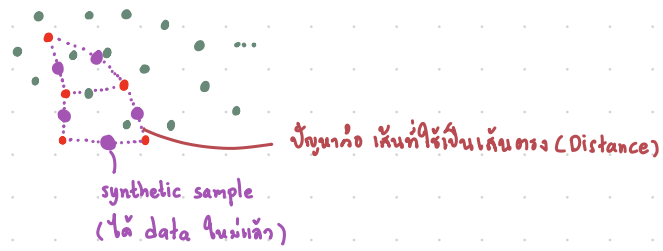
① Under sampling



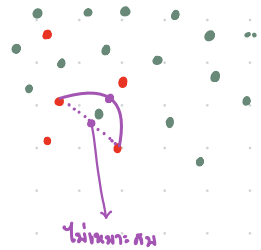
② Oversampling ; เลือกตัวที่น้อย แล้วเพิ่มเข้าไป



③ Data Augmentation (SMOTE: Synthetic Minority Over-sampling Technique (2002))



How? จุด • ออกมาข้างนอก
How? เสนาใจออก majority (กลุ่มใหญ่)

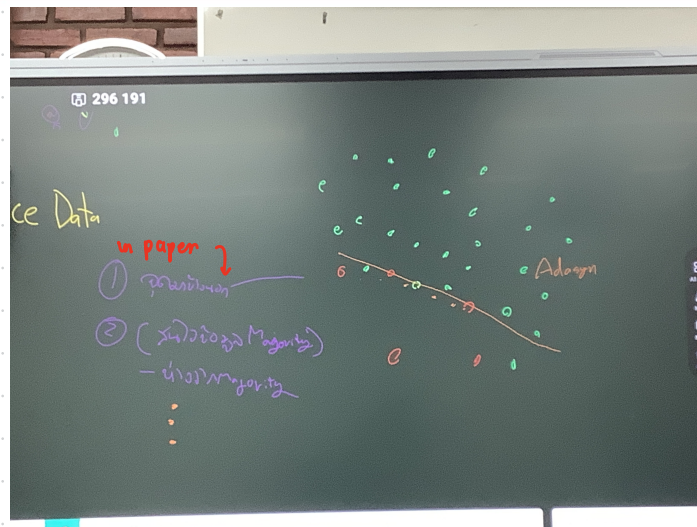


③ Data Augmentation (ADASYN)



READ THE RIGHT-HAND PANEL

Only 17 of 46 minority points have at least half their neighbours in the majority class, yet ADASYN gives them 59% of the budget. SMOTE would give the same group 37%. Interior points shrink to nothing; the boundary swells.



The data for the plot in Figure 3 was extracted from a Mammography dataset¹ (Woods et al., 1993). The minority class samples are shown by + and the majority class samples are shown by o in the plot. In Figure 3(a), the region indicated by the solid-line rectangle is a majority class decision region. Nevertheless, it contains three minority class samples shown by '+' as false negatives. If we replicate the minority class, the decision region for the minority class becomes very specific and will cause new splits in the decision tree. This will lead to more terminal nodes (leaves) as the learning algorithm tries to learn more and more specific regions of the minority class; in essence, overfitting. Replication of the minority class does not cause its decision boundary to spread into the majority class region. Thus, in Figure 3(b), the three samples previously in the majority class decision region now have very specific decision regions.

4.2 SMOTE

We propose an over-sampling approach in which the minority class is over-sampled by creating “synthetic” examples rather than by over-sampling with replacement. This approach is inspired by a technique that proved successful in handwritten character recognition (Ha & Bunke, 1997). They created extra training data by performing certain operations on real data. In their case, operations like rotation and skew were natural ways to perturb the training data. We generate synthetic examples in a less application-specific manner, by operating in “feature space” rather than “data space”. The minority class is over-sampled by taking each minority class sample and introducing synthetic examples along the line segments joining any/all of the k minority class nearest neighbors. Depending upon the amount of over-sampling required, neighbors from the k nearest neighbors are randomly chosen. Our implementation currently uses five nearest neighbors. For instance, if the amount of over-sampling needed is 200%, only two neighbors from the five nearest neighbors are chosen and one sample is generated in the direction of each. Synthetic samples are generated in the following way: Take the difference between the feature vector (sample) under consideration and its nearest neighbor. Multiply this difference by a random number between 0 and 1, and add it to the feature vector under consideration. This causes the selection of a random point along the line segment between two specific features. This approach effectively forces the decision region of the minority class to become more general.

Algorithm *SMOTE*, on the next page, is the pseudo-code for SMOTE. Table 4.2 shows an example of calculation of random synthetic samples. The amount of over-sampling is a parameter of the system, and a series of ROC curves can be generated for different populations and ROC analysis performed.

The synthetic examples cause the classifier to create larger and less specific decision regions as shown by the dashed lines in Figure 3(c), rather than smaller and more specific regions. More general regions are now learned for the minority class samples rather than those being subsumed by the majority class samples around them. The effect is that decision trees generalize better. Figures 4 and 5 compare the minority over-sampling with replacement and SMOTE. The experiments were conducted on the mammography dataset. There were 10923 examples in the majority class and 260 examples in the minority class originally. We have approximately 9831 examples in the majority class and 233 examples

1. The data is available from the USF Intelligent Systems Lab, <http://morden.csee.usf.edu/~chawla>.